IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO CRESCENT AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

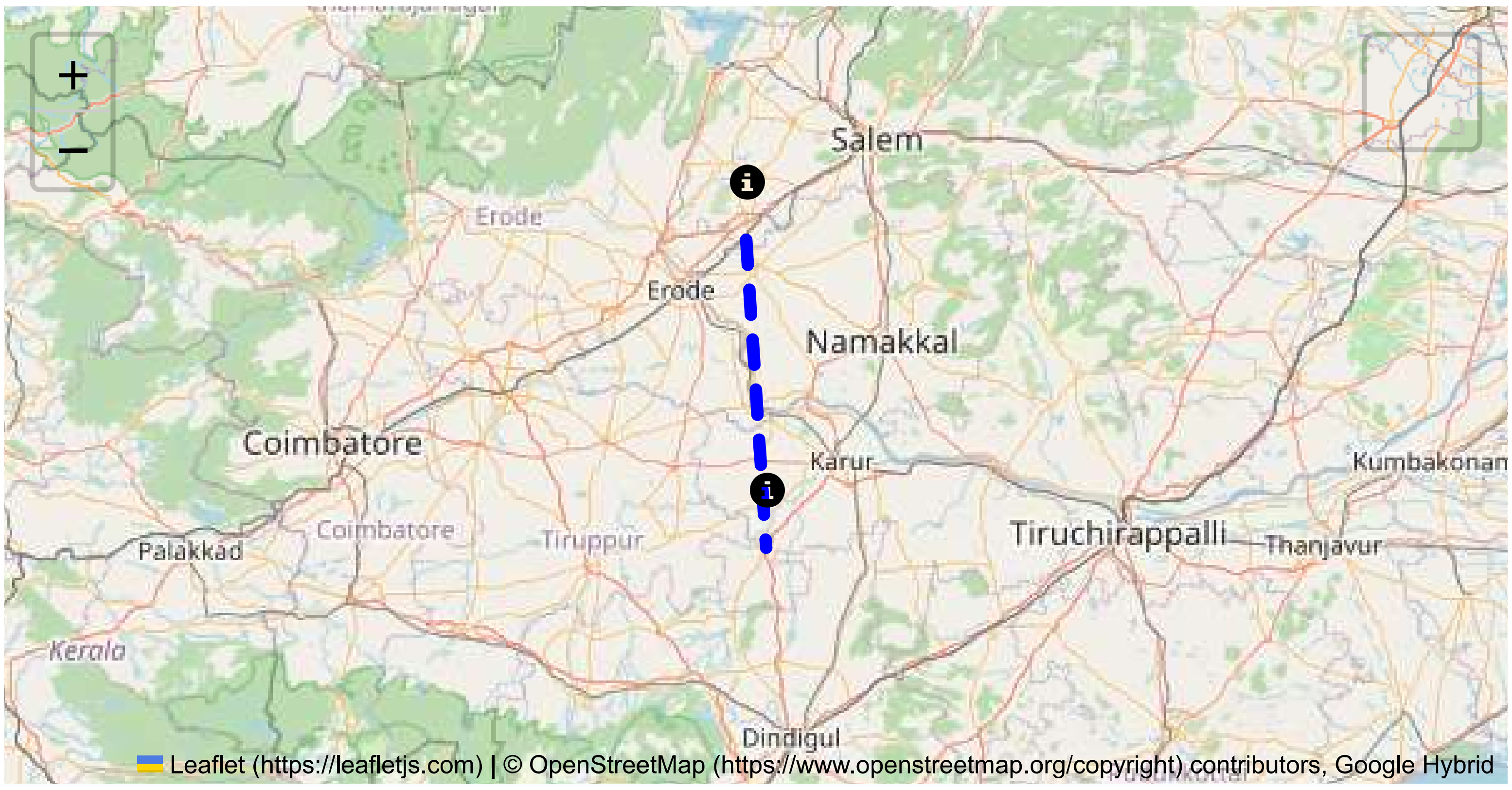
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

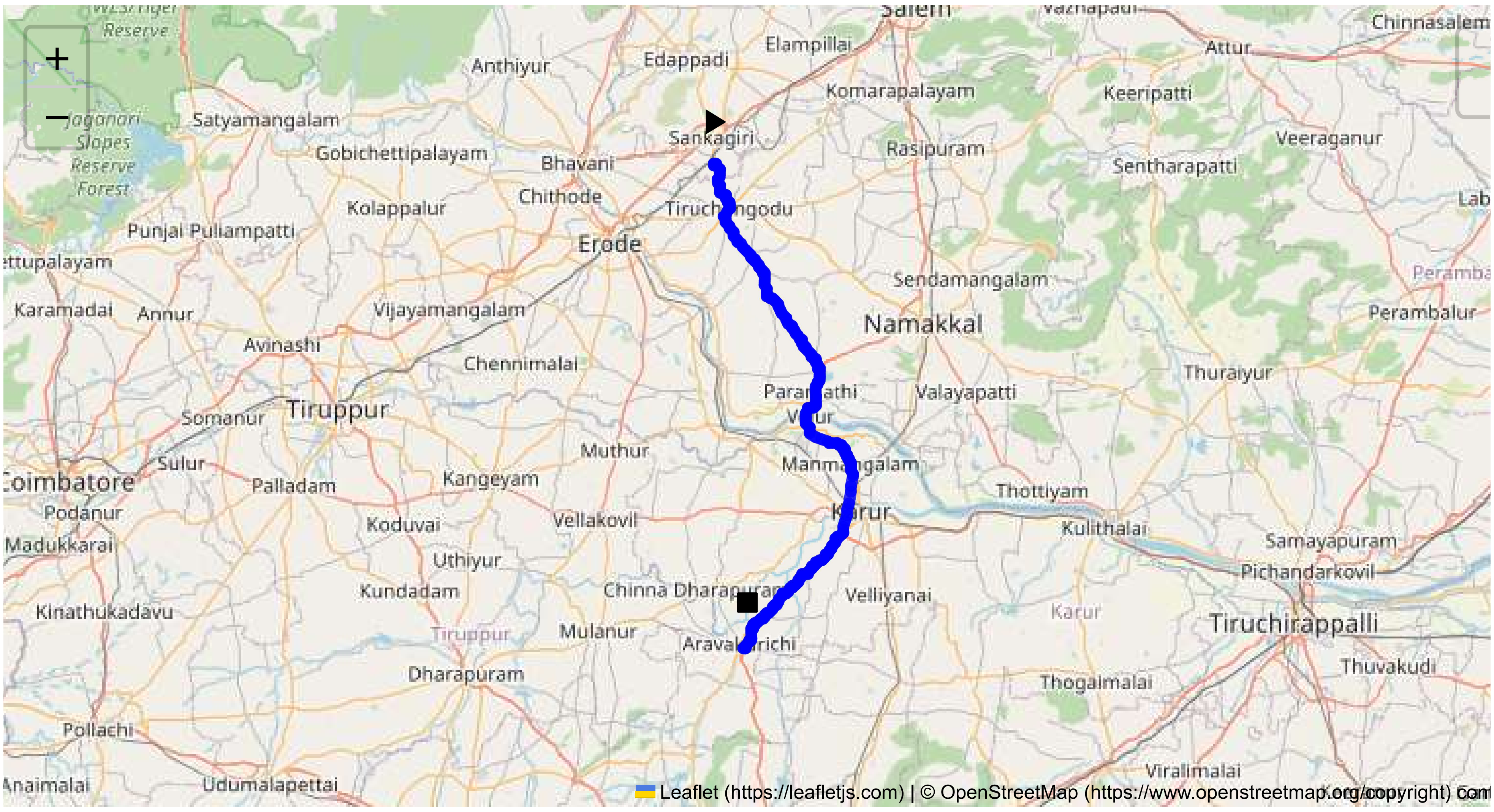
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 92.38 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.1 hours
Start: (11.4381, 77.8734)
End: (10.772697, 77.916647)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Aravakurichi covers approximately 92.38 kilometers, generally following National highways and state routes. Given the region's geography, the drive predominantly travels through rural and semi-urban areas, interspersed with small towns and agricultural land. The primary route follows NH44 and connects with local roads leading to Aravakurichi.

Typical Weather Conditions

- **Climate**: Tamil Nadu typically experiences a tropical climate, characterized by hot summers (March to June), monsoon rains (June to September), and mild winters (October to February).
- **Weather-related Hazards**: During the monsoon, heavy rain can lead to flooding, affecting road conditions and visibility. In summers, heat can cause road surface damage or contribute to driver fatigue.

Traffic Patterns

- **Peak Hours**: Traffic congestion in Tamil Nadu peaks during morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 7:00 PM) hours, especially near major towns.
- **Congestion-prone Areas**: Potential bottlenecks include town centers and intersections in major towns such as Erode and Karur, where local traffic, markets, and bus stations may cause delays.

Road Quality and Infrastructure

- **Road Condition**: NH44 is generally well-maintained, with periodic potholes in local segments and rural roads. Road markings and signage can sometimes be sparse or poorly visible, particularly after rains or at night.
- **Infrastructure**: Bridges and crossings are in good condition, though some may have narrow lanes unsuitable for wide vehicles.

Alternative Routes for Emergencies

- Possible detours are limited due to the rural landscape. However, alternative state highways can be used subject to road conditions, but they often lead to longer travel times.
- Local roads may offer bypass options around congested towns but require awareness of their current conditions.

Local Regulations on Hazardous Material Transport

- Transporting hazardous materials requires adherence to national safety standards, including using marked vehicles and carrying documentation such as Material Safety Data Sheets (MSDS).
- There are likely restrictions on the movement of such materials during night-time and local festivals or events.

Historical Incidents Involving Heavy Vehicles

- **Accident Data**: There is limited public data on history accidents along this specific route. However, state highways in Tamil Nadu have witnessed incidents due to over-speeding, driver fatigue, and vehicle overloading.
- Local reports indicate occasional incidents involving heavy vehicles mainly at intersections and during adverse weather.

Environmental Considerations and Sensitive Areas

- **Protected Areas**: Although the route does not pass through nationally protected wildlife reserves, awareness of local wildlife crossings is necessary.
- **Agricultural Zones**: There are areas of prime agricultural land where accidents involving hazardous materials could cause environmental damage.

Communication Coverage

- **Network Providers**: Major network providers cover Tamil Nadu, though there may be signal drops in remote or heavily forested areas.
- **Dead Zones**: Expect potential signal loss primarily in rural stretches and under significant weather conditions.

Emergency Response Times

- **Urban Areas**: Response times in towns like Salem or Karur may vary from 15 to 30 minutes, given the availability of local hospitals and emergency services.
- **Rural Segments**: Further from towns, response times may extend to 60 minutes or more, given the rural landscape and road accessibility challenges.

Overall Risk Assessment Summary

- **Route Challenges**: Major challenges include weather-related disruptions, occasional traffic congestion, and variable road conditions. Main safety concerns relate to monsoon impacts and potential for agricultural damage.
- **Preparation Recommendations**: Ensure drivers are prepared for weather conditions, have redundant communication options, and are familiar with alternative routes and the nearest emergency services.

This assessment highlights the importance of thorough preparation and situational awareness when transporting hazardous materials in the region. It underlines the necessity for continuous monitoring and contingency planning to mitigate potential risks along the route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44019, 77.87547	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
4	Turn	Medium	11.39685, 77.88861	30 KM/Hr
5	Turn	High	11.37862, 77.89488	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16755, 78.01568	30 KM/Hr
8	Turn	Medium	11.15579, 78.02169	30 KM/Hr
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Turn	High	11.14485, 78.02282	15 KM/Hr
11	Turn	High	10.77130, 77.91658	15 KM/Hr
12	Blind Spot	Blind Spot	10.77149, 77.91619	10 KM/Hr
13	Blind Spot	Blind Spot	10.77259, 77.91697	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium
19	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium
20	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
21	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots